

AB RENEWABLE ENERGY ENGINEERING

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BIOGAS DIGESTION

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1. Gas consisting mainly of methane and carbon dioxide that produces putrefactive bacteria that breaks down organic material under airless-condition.
 - a. Biofuel
 - ☒ b. Biogas
 - c. Gasifier
 - d. All of the above
2. Primary gas produced during anaerobic digestion of agricultural wastes.
 - a. CH_4 & H_2S
 - ☒ b. CH_4 & CO_2
 - c. CH_4 & CO
 - d. All of the above
3. Gases produced from biogas.
 - ☒ a. CO_2 and CH_4
 - b. CO , H_2 , and CH_4
 - c. CO_2
 - d. All of the above
 - e. None of the above
4. Biogas is generated inside the _____.
 - a. gas holder
 - ☒ b. mixing tank
 - c. digester
 - d. None of the above
5. Gas produced from animal residues consisting of 95% CH_4 and 5% CO_2 .
 - a. Biogas
 - b. Producer gas
 - ☒ c. Synthetic gas
 - d. None of the above

Natural gas
6. Feed-material-to-water ratio for optimum biogas generation is _____.
 - ☒ a. 1:0.5 to 1:1
 - b. 1:1 to 1:2
 - c. 1:2 to 1:3
 - d. None of the above

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7. Biogas generation is a process of _____.
 - ☒ a. anaerobic digestion
 - b. anaerobic gasification
 - c. fermentation of yeast
 - d. None of the above
8. C-N ratio suitable for anaerobic digestion of agriwastes.
 - ☒ a. 10:1 to 20:1
 - b. 20:1 to 30:1
 - c. 30:1 to 40:1
 - d. None of the above
9. Substrate for biogas production.
 - a. Poultry manure
 - b. Piggery manure
 - c. Cattle manure
 - ☒ d. Agri-plant residue
 - e. All of the above
 - f. None of the above
10. Residue that comes out of the digester of a biogas plant after the substrates are digested.
 - a. Sludge
 - ☒ b. Slurry
 - c. Effluent
 - d. None of the above
11. Layer of floating fibrous material on the slurry.
 - ☒ a. Scum
 - b. Substrate
 - c. Sludge
 - d. None of the above
12. Optimum hydraulic retention time for biogas production.
 - ☒ a. 15 – 20 days
 - b. 20 – 25 days
 - c. 25 to 30 days
 - d. None of the above

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13. Retention time of pig manure in a biogas digester is _____ than that of chicken manure.

- a. shorter
- b. the same
- c. longer
- d. None of the above

14. Using biogas as fuel for engine can replace _____ of diesel fuel.

- a. 60%
- b. 80%
- c. 100%
- d. None of the above

15. A biogas digester is to be designed to accommodate 30 liters of dung per day. If the feed-material-to-water ratio is 1:1 and the designed retention time is 80 days, what is the capacity of the digester?

- a. 4800 liters
- b. 5200 liters
- c. 6100 liters
- d. None of the above

16. A tunnel-ventilated poultry farm is composed of 5 buildings with 30,000 birds per building. How many kilos of manure is available in each building if the birds will stay in there for 30 days? Consider a manure yield of 0.025 kg/day/bird.

- a. 18,500 kg
- b. 20,875 kg
- c. 22,500 kg
- d. None of the above

17. Biogas can be used for _____.

- a. cooking
- b. lighting
- c. engine fuel
- d. heating refrigerant
- e. All of the above
- f. None of the above

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18. Heating value of biogas is around _____.

- a. 5,500 kcal/m³
- b. 7,200 kcal/m³
- c. 9,468 kcal/m³
- d. None of the above

19. Retention time of animal manure mixed with plant materials inside the digester is _____ than that with no plant materials.

- a. shorter
- b. the same
- c. longer
- d. None of the above

20. When biogas contains 40% methane and 60% carbon dioxide, it can _____.

- a. be used for cooking
- b. be used for lighting
- c. run a gas refrigerator
- d. None of the above

21. Which of the following statements is true?

- a. The methane content of biogas is higher than that of the natural gas.
- b. The methane content of biogas is lower than that of the natural gas.
- c. The methane content of biogas is the same with that of the natural gas.
- d. All of the above
- e. None of the above

22. Biogas unit designed to use human wastes as the principal organic input.

- a. Bio-digester
- b. Bio-latrline
- c. Bio-fermentor
- d. None of the above

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GOD BLESS !!!

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